

VULNERABILITY ASSESSMENT REPORT

Target Website:

<https://www.tcs.com>

Prepared By:

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Task 1

OBJECTIVE

The objective of this assessment is to perform a passive security analysis of the TCS website using browser developer tools and Nmap.

TOOLS USED

- Browser developer tools
- Nmap
- Canva

METHODOLOGY

1. Accessed the target website (www.tcs.com).
2. Verified secure HTTPS communication.
3. Examined HTTP security headers using Browser Developer Tools.
4. Conducted network reconnaissance using Nmap.
5. Recorded and analyzed all observations.

FINDINGS

1. The target website was observed to be using HTTPS, which helps ensure that communication between users and the website remains encrypted and secure.
2. During the header analysis, multiple security headers were identified, indicating that the website has implemented measures to protect against common web-based attacks.
3. Content Security Policy (CSP) was present, helping to reduce the risk of malicious script execution and content injection attacks.
4. Strict Transport Security (HSTS) was configured, ensuring that browsers communicate with the website only through secure HTTPS connections.
5. Additional security headers such as X-Frame-Options and X-Content-Type-Options were observed, providing protection against clickjacking and MIME-type related attacks.
6. The Nmap scan revealed that only ports 80 and 443 were accessible, which is expected for a public-facing web application.
7. Most other TCP ports were found to be filtered, limiting unnecessary network exposure and reducing the potential attack surface.
8. Based on the passive assessment conducted, no critical vulnerabilities or high-risk security issues were identified on the target website.

RISK ASSESSMENT

Overall Risk Level: Low

The website demonstrates the implementation of several recommended security practices, including HTTPS enforcement, security header configuration, and limited network exposure. No critical vulnerabilities were identified during the passive assessment.

EVIDENCE 1: BROWSER ANALYSIS

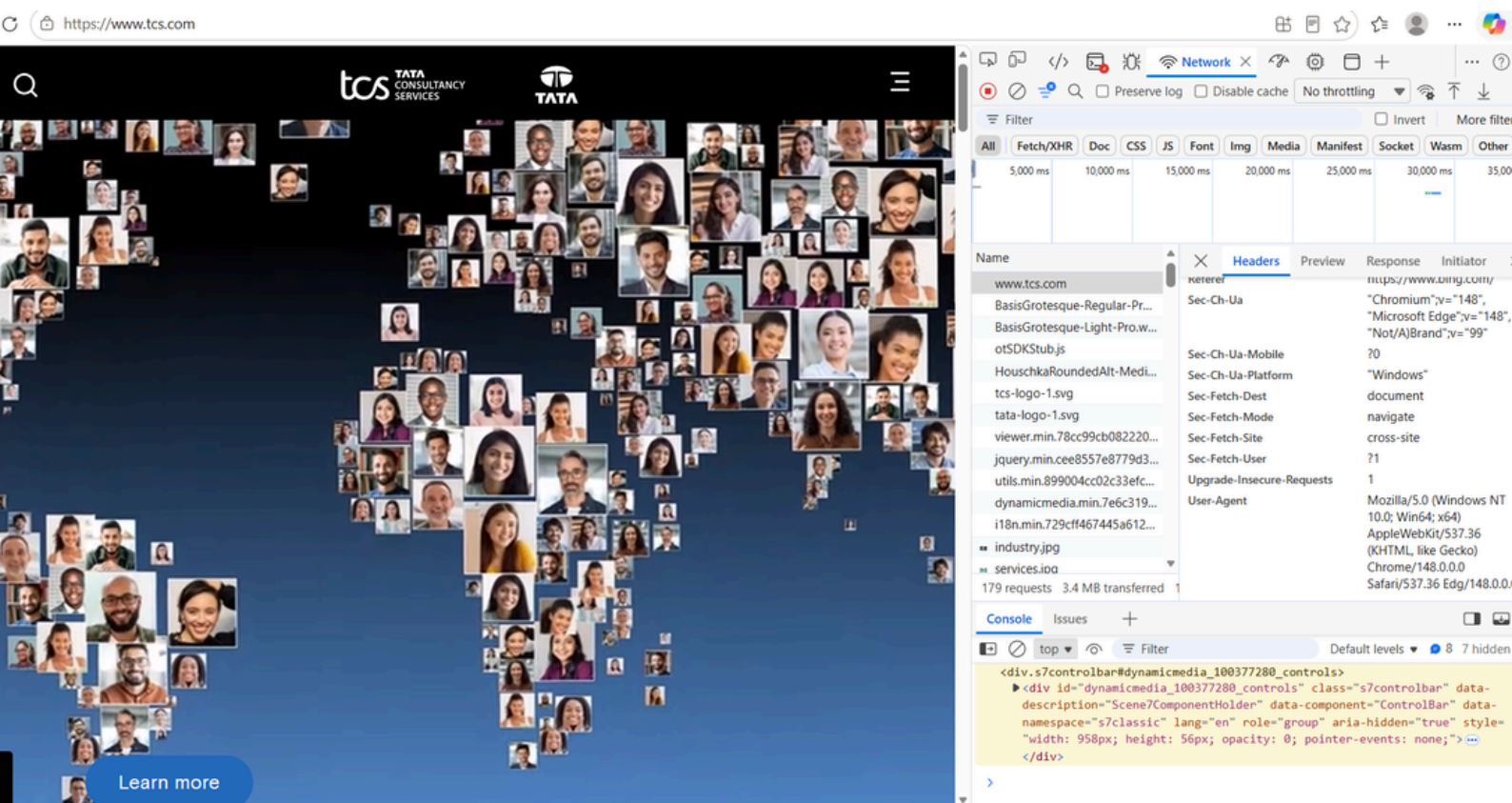
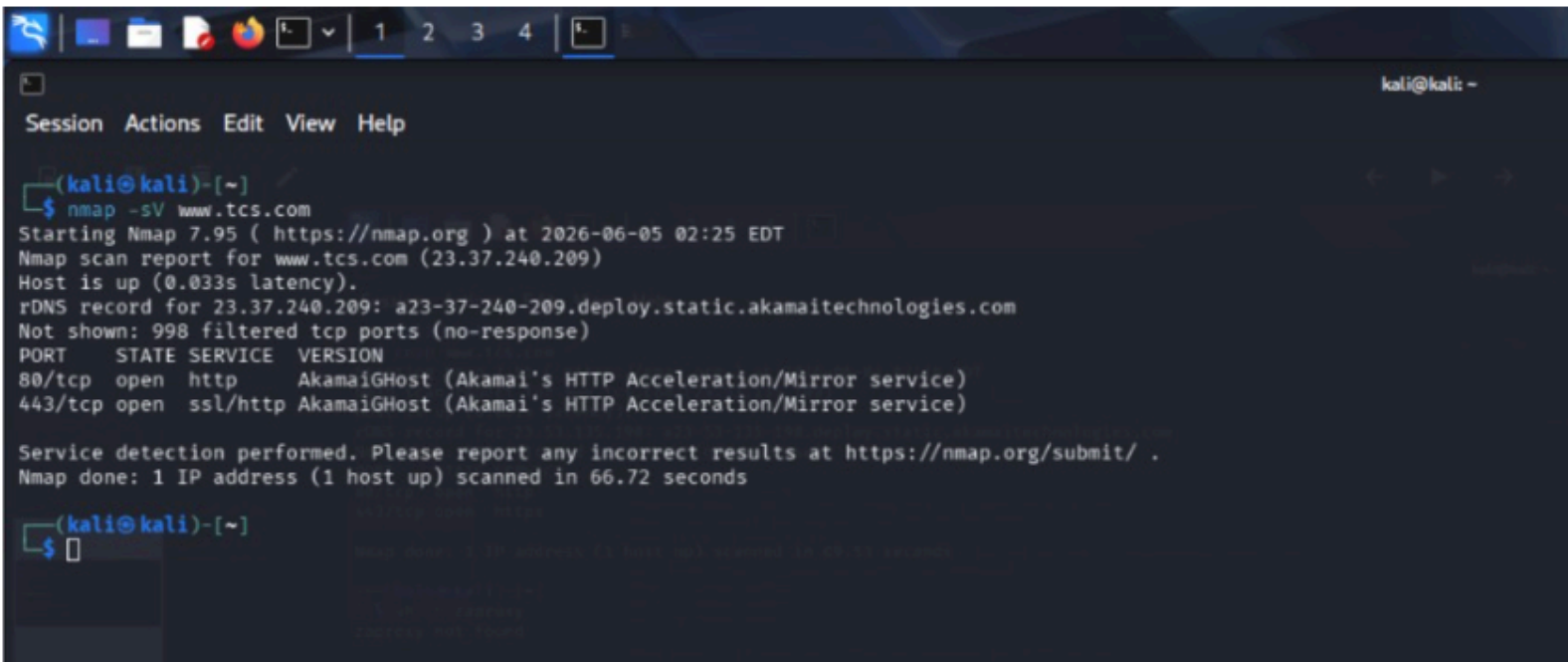


Figure 1: Browser Developer Tools showing HTTPS implementation and security-related response headers on the TCS website.

EVIDENCE 2: NMAP SCAN RESULTS



```
(kali@kali)~  
$ nmap -sV www.tcs.com  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-05 02:25 EDT  
Nmap scan report for www.tcs.com (23.37.240.209)  
Host is up (0.033s latency).  
rDNS record for 23.37.240.209: a23-37-240-209.deploy.static.akamaitechnologies.com  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE VERSION  
80/tcp    open  http    AkamaiGHost (Akamai's HTTP Acceleration/Mirror service)  
443/tcp    open  ssl/http AkamaiGHost (Akamai's HTTP Acceleration/Mirror service)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 66.72 seconds  
  
(kali@kali)~  
$
```

Figure 2: Nmap service detection scan performed on www.tcs.com. The scan identified ports 80 (HTTP) and 443 (HTTPS) as accessible, while most other TCP ports were filtered, indicating limited network exposure.

OBSERVATION

The Nmap service detection scan identified ports 80 and 443 as accessible on the target website. These ports are commonly used for HTTP and HTTPS communication. Most other TCP ports were filtered, which helps reduce unnecessary network exposure and limits potential attack vectors. This indicates that only essential web services are publicly accessible.

RECOMENDATIONS

1. Continue using HTTPS to ensure secure communication between users and the website.
2. Review security headers regularly to maintain protection against common web threats.
3. Limit unnecessary server information disclosure wherever possible.
4. Perform periodic security assessments to identify and address potential risks.
5. Apply security updates and monitor the website regularly to maintain a strong security posture.

CONCLUSION

The vulnerability assessment of the TCS website was carried out using Browser Developer Tools and Nmap. The analysis confirmed that HTTPS was properly implemented, important security headers were configured, and network exposure was limited to essential web services. No critical vulnerabilities were identified during the passive assessment. Overall, the website demonstrates good security practices and maintains a strong security posture.

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